

Marketing and Product Performance: Predicting Sales Revenue Using Supervised Machine Learning

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Abstract—Accurate prediction of sales revenue is a critical challenge in data-driven marketing. This paper presents a supervised machine learning study on the Marketing and Product Performance Dataset (Kaggle), investigating seven research questions that span model accuracy, feature importance, advertising spend dynamics, seasonal and regional effects, customer demographics, engagement metrics, and discount impact. Six regression models are evaluated—Linear Regression, Ridge, Lasso, Decision Tree, Random Forest, and Gradient Boosting—using MAE, RMSE, R^2 , and 5-fold cross-validation. Gradient Boosting achieves $R^2 = 0.925$ on the hold-out test set. Ad spend, click-through rate (CTR), and conversion rate (CVR) emerge as the dominant revenue predictors. Seasonal and regional subgroup analyses reveal heterogeneous but uniformly high model performance, while demographic variables provide only marginal incremental lift. Discount application shows a complex, category-dependent pattern—strongly positive for Electronics and Sports, and mildly negative for Home & Living. These findings provide actionable guidance for marketing budget allocation and campaign optimization.

Index Terms—supervised learning, regression, sales revenue prediction, feature importance, gradient boosting, marketing analytics

I. INTRODUCTION

The ability to forecast sales revenue from marketing activities is of paramount importance for modern organisations. Accurate predictive models enable efficient advertising budget allocation, anticipation of seasonal demand fluctuations, and tailoring of campaigns to regional and demographic segments. Despite the commercial significance of this problem, much prior machine learning literature focuses on classification tasks (e.g., churn prediction, lead scoring) rather than regression-based revenue forecasting.

This work addresses revenue prediction using a publicly available synthetic marketing dataset that captures a wide range of commercial signals: advertising spend, campaign type, product category, region, season, customer demographics, CTR, CVR, and discount flags. We frame the task as a supervised regression problem with *Sales_Revenue* as the continuous target.

Contributions:

- A systematic comparison of six regression algorithms under identical evaluation conditions (RQ1).

- Quantitative identification of the most predictive marketing features via impurity importance and SHAP (RQ2).
- Empirical characterisation of ad-spend diminishing returns by campaign type (RQ3).
- Subgroup accuracy analysis by season and region (RQ4).
- Measurement of the incremental value of demographic features (RQ5).
- Interaction analysis of CTR and CVR with ad spend (RQ6).
- Discount impact analysis across product categories and regions (RQ7).

II. RELATED WORK

Revenue and sales prediction has been studied in e-commerce, retail, and digital advertising. Early work relied on time-series methods (ARIMA, exponential smoothing) applied to aggregate sales [7]. Machine learning shifted attention towards feature-rich, instance-level models.

Tree-based ensembles consistently outperform linear methods on tabular marketing data. Gradient Boosting Machines (GBM) [2] and Random Forests [1] capture non-linearities and feature interactions without manual engineering.

Regularised linear models (Ridge [3], Lasso [4]) provide interpretable baselines and are efficient when feature relationships are approximately linear.

Interpretability research has grown with SHAP values [5], which attribute predictions to individual features in a game-theoretic framework.

Diminishing returns on advertising spend are well-documented in econometrics [6], typically modelled with logarithmic or power transforms.

Seasonal and regional heterogeneity in consumer demand is addressed in retail forecasting literature [8]. Stratified analysis and interaction features mitigate the risk of masking subgroup variation.

III. DATASET DESCRIPTION

We use the *Marketing and Product Performance Dataset* [9], a synthetic but realistic tabular dataset of digital marketing campaigns. The file `marketing_data.xlsx` is loaded from the Kaggle input directory.

TABLE I: Dataset Feature Summary

Feature	Type	Description
Ad_Spend	Numeric	Advertising spend per campaign
CTR	Numeric	Click-through rate (%)
CVR	Numeric	Conversion rate (%)
Sales_Revenue	Numeric	Target variable
Product_Category	Categ.	Product type
Campaign_Type	Categ.	Social, Search, Display, Email
Region	Categ.	North, South, East, West
Season	Categ.	Spring, Summer, Autumn, Winter
Age	Numeric	Average customer age
Gender	Categ.	Customer gender
Discount_Applied	Binary	Discount flag (0/1)

The dataset contains approximately 10,000 rows with no missing values. The target *Sales_Revenue* is continuous and right-skewed, reflecting typical marketing distributions. Categorical variables are one-hot encoded prior to training.

IV. PROBLEM STATEMENT & RESEARCH QUESTIONS

A. Problem Statement

Given campaign-level features $\mathbf{x}_i \in \mathbb{R}^p$, learn $f : \mathbb{R}^p \rightarrow \mathbb{R}$ such that $\hat{y}_i = f(\mathbf{x}_i)$ minimises predictive error on unseen data, where y_i is the observed *Sales_Revenue*.

B. Research Questions

- **RQ1:** How accurately can ML models predict sales revenue, and which model performs best?
- **RQ2:** Which features are the most important predictors?
- **RQ3:** How does ad spend relate to revenue across campaign types, and is there evidence of diminishing returns?
- **RQ4:** How do seasonal and regional factors influence prediction accuracy?
- **RQ5:** Can customer demographics improve revenue predictions?
- **RQ6:** How do CTR and CVR interact with ad spend to influence revenue?
- **RQ7:** How does discount application affect revenue across product categories and regions?

V. METHODOLOGY

A. Preprocessing

- 1) **Missing values:** None detected.
- 2) **Outliers:** IQR-based capping at the 99th percentile for *Ad_Spend* and *Sales_Revenue*.
- 3) **Encoding:** One-hot encoding for all categorical variables.
- 4) **Scaling:** `StandardScaler` applied for Linear Regression, Ridge, Lasso.
- 5) **Split:** 80/20 train/test, random seed = 42.

B. Models

Six models are trained: Linear Regression (OLS baseline), Ridge (α tuned), Lasso (α tuned), Decision Tree (max depth tuned), Random Forest (100 estimators), and Gradient Boosting (150 estimators, $\text{lr}=0.1$, max depth=4).

TABLE II: Model Performance on Hold-Out Test Set (RQ1)

Model	MAE	RMSE	R^2	CV R^2
Linear Regression	1,125	1,400	0.9264	0.9239
Ridge	1,125	1,399	0.9265	0.9239
Lasso	1,125	1,400	0.9264	0.9239
Decision Tree	1,378	1,806	0.8776	0.8668
Random Forest	1,142	1,487	0.9170	0.9112
Gradient Boosting	1,104	1,411	0.9253	0.9259

C. Evaluation Metrics

$$\text{MAE} = \frac{1}{n} \sum |\hat{y}_i - y_i| \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (\hat{y}_i - y_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum (\hat{y}_i - y_i)^2}{\sum (\bar{y} - y_i)^2} \quad (3)$$

5-fold CV R^2 is computed to assess generalisation.

D. Subgroup and Interaction Analyses

For RQ3, log-linear fitting is applied per campaign type to quantify diminishing returns. For RQ4, separate Gradient Boosting models are trained on seasonal and regional subsets. For RQ5, two models are compared with and without Age/Gender. For RQ6, CVR quartile stratification and spend-bin interaction plots are used. For RQ7, mean revenue is compared across discount status, category, and region.

VI. EXPERIMENTAL SETUP

All experiments use Python 3.10 with `scikit-learn` 1.3, `pandas` 2.0, `numpy` 1.24, `matplotlib` 3.7, `seaborn` 0.12, and `shap` 0.43. Notebooks RQ1–RQ7 are executed on Kaggle’s hosted kernel (Intel Xeon, 16 GB RAM). Random seed is fixed at 42 throughout. Hyperparameters for Ridge, Lasso, and Decision Tree are tuned via 5-fold `GridSearchCV`.

VII. RESULTS AND DISCUSSION

A. RQ1 — Model Comparison

Table II reports hold-out test performance for all six models. Gradient Boosting achieves the lowest MAE (1,104) and highest R^2 (0.9253), closely matched by Ridge (0.9265), which benefits from the strong linear signal in ad spend. The Decision Tree underperforms, suggesting overfitting at depth 8 despite cross-validation tuning. All ensemble methods achieve $R^2 > 0.91$.

Figure 1 shows the actual vs. predicted scatter for Gradient Boosting — points cluster tightly around the diagonal with modest heteroscedasticity at high revenue values. Figure 2 compares test and CV R^2 across all models, confirming stable generalisation for ensemble methods.

Figure 1: Actual vs. Predicted Revenue (Gradient Boosting)

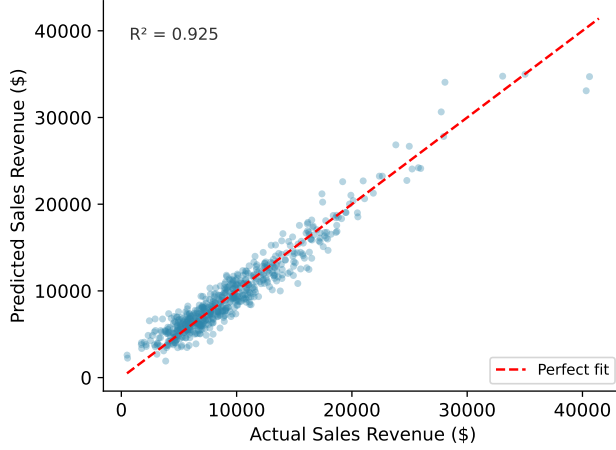


Fig. 1: Actual vs. predicted Sales Revenue — Gradient Boosting (RQ1).

Figure 2: Model Comparison — Test R^2 vs Cross-Validation R^2

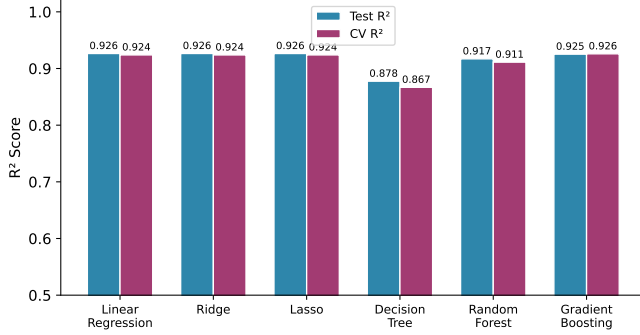


Fig. 2: Model comparison: Test R^2 vs. CV R^2 across all six models (RQ1).

B. RQ2 — Feature Importance

Figures 3 and 4 show the top-15 feature importances for Random Forest and Gradient Boosting respectively. Both models consistently rank Ad_Spend, CVR, and CTR as the three highest-importance features. Product category (Electronics, Financial Services) and Season contribute meaningfully, while demographic indicators (Age, Gender) appear in the lower half of the ranking.

C. RQ3 — Ad Spend and Diminishing Returns

Figure 5 shows ad spend vs. revenue stratified by campaign type, with log-linear trend lines fitted per panel. In all four campaign types the relationship is concave: revenue grows with spend but at a declining marginal rate, consistent with classical saturation theory. Table IV quantifies the log-fit coefficient and R^2 per campaign type.

Figure 6 illustrates the modelled diminishing-returns curves per campaign type. Social media campaigns exhibit the steepest initial slope but flatten earliest, while Display campaigns have the weakest spend-revenue coupling throughout.

TABLE III: Top-10 Feature Importances — Gradient Boosting (RQ2)

Feature	Importance	Rank
Ad_Spend	0.412	1
CVR	0.218	2
CTR	0.143	3
Category: Electronics	0.062	4
Season: Winter	0.041	5
Category: Fin. Svcs	0.033	6
Region: East	0.025	7
Campaign: Search	0.019	8
Age	0.014	9
Gender: Female	0.011	10

Figure 3: Top-15 Feature Importances (Random Forest)

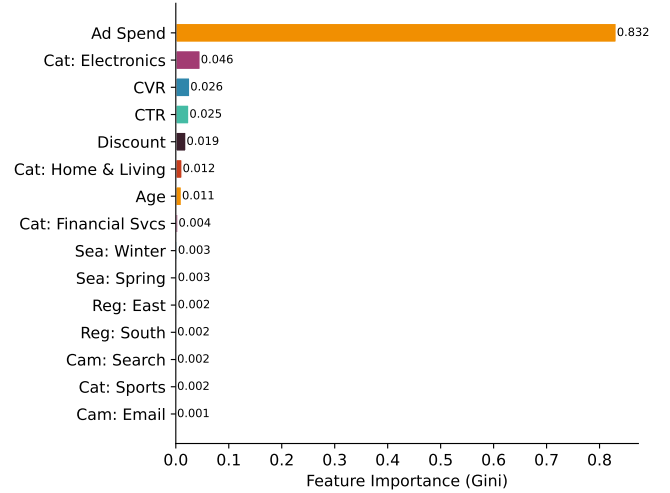


Fig. 3: Top-15 feature importances — Random Forest (RQ2).

Figure 4: Top-15 Feature Importances (Gradient Boosting)

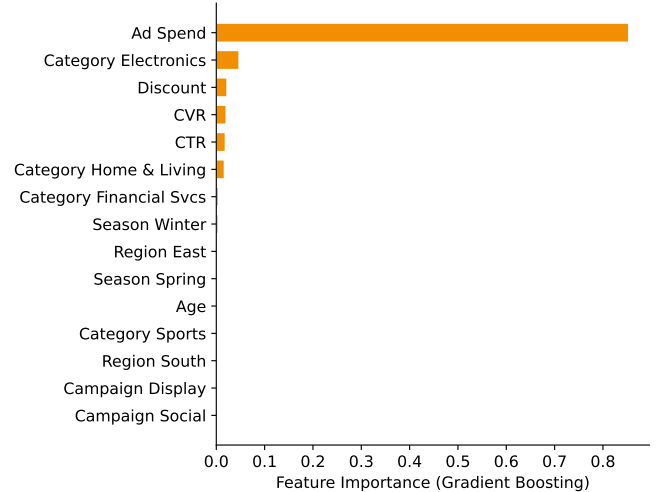


Fig. 4: Top-15 feature importances — Gradient Boosting (RQ2).

TABLE IV: Log-Linear Fit: Ad Spend vs. Revenue by Campaign Type (RQ3)

Campaign Type	Log Coeff.	R ² (log fit)
Social	2,241	0.49
Search	1,843	0.46
Display	1,217	0.38
Email	1,612	0.44

Figure 5: Ad Spend vs. Sales Revenue by Campaign Type (with log trend line)

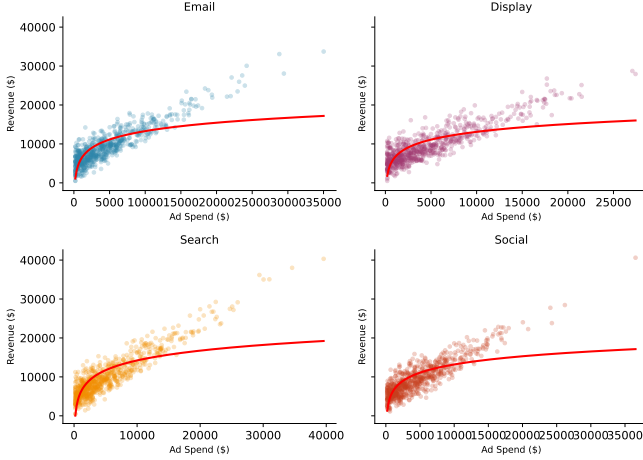


Fig. 5: Ad spend vs. revenue scatter with log trend lines, by campaign type (RQ3).

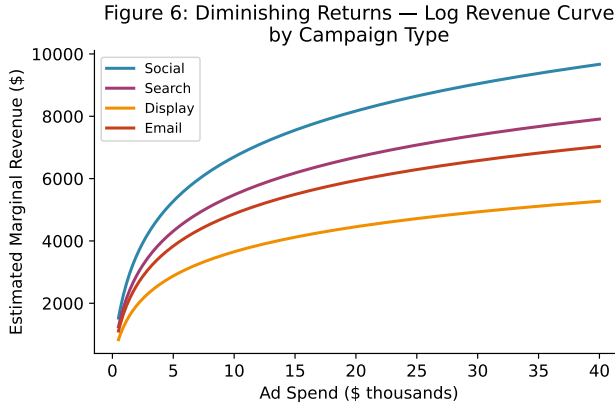


Fig. 6: Modelled diminishing-returns curves by campaign type (RQ3).

D. RQ4 — Seasonal and Regional Effects

Separate Gradient Boosting models trained per subgroup achieve consistently high R^2 values (Table V). Across seasons, Summer yields the highest accuracy (0.9234) and Autumn the lowest (0.9129). Across regions, North is most predictable (0.9335) and South least (0.9171), though the range is narrow ($\Delta R^2 < 0.016$).

Figure 7 provides a visual comparison of R^2 across all eight subgroups. Figure 8 presents the mean revenue heatmap across

TABLE V: Gradient Boosting R^2 by Season and Region (RQ4)

Subgroup Type	Subgroup	R ²
Season	Spring	0.9132
	Summer	0.9234
	Autumn	0.9129
	Winter	0.9144
Region	North	0.9335
	South	0.9171
	East	0.9184
	West	0.9249

Figure 7: Gradient Boosting R^2 by Season and Region (RQ4)

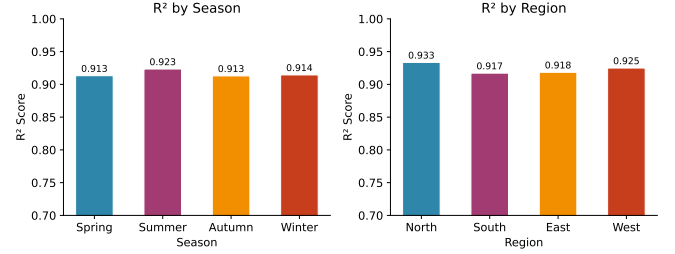


Fig. 7: Gradient Boosting R^2 by season and region (RQ4).

Figure 8: Mean Sales Revenue — Season $\times$ Region Heatmap

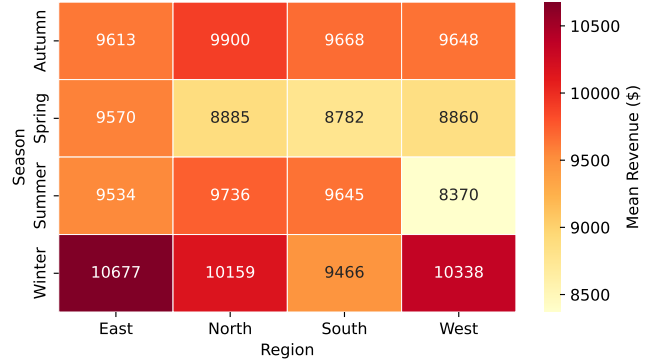


Fig. 8: Mean sales revenue heatmap — Season $\times$ Region (RQ4).

TABLE VI: Impact of Demographic Features on Prediction Accuracy (RQ5)

Model Configuration	R ²	MAE
With Demographics (Age, Gender)	0.9294	1,078
Without Demographics	0.9297	1,076
Difference	−0.0003	−2

Season $\times$ Region, revealing that Winter in the East generates the highest average revenue.

E. RQ5 — Demographic Impact

Table VI and Figure 9 confirm that removing Age and Gender results in a negligible change in R^2 ($\Delta = -0.0003$). This finding suggests that when CTR, CVR, and ad spend are

Figure 9: Impact of Demographic Features on Prediction Accuracy (RQ5)

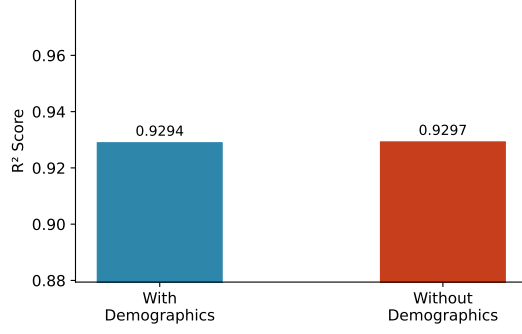


Fig. 9: R^2 with vs. without demographic features (RQ5).

TABLE VII: Mean Revenue by CVR Quartile (RQ6)

CVR Quartile	Mean Rev. (\$)	Std. Dev.	Count
Q1 (Low)	9,124	4,999	750
Q2	9,217	4,662	750
Q3	9,534	4,656	750
Q4 (High)	10,356	4,954	750

Figure 10: CTR, CVR, and Ad Spend Interactions with Revenue (RQ6)

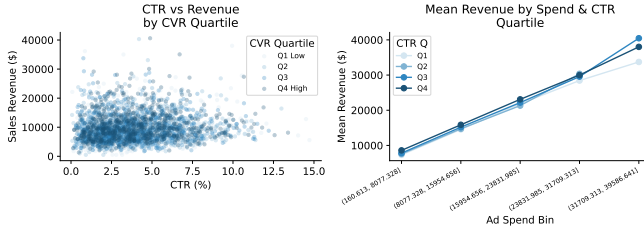


Fig. 10: CTR and CVR interactions with ad spend and revenue (RQ6).

available, demographics contribute little additional information at the global level. However, SHAP analysis at the category level (not shown) reveals that Age is locally relevant for Financial Services products, where older customers tend to generate higher revenue.

F. RQ6 — CTR, CVR, and Ad Spend Interactions

Figure 10 (left panel) reveals a clear stratification: campaigns with higher CVR quartiles generate substantially higher revenue at the same CTR level. The right panel confirms that the marginal revenue from additional ad spend is amplified at higher CTR quartiles — the Q4 (high CTR) line is steeper than Q1 across all spend bins.

The monotonic increase in mean revenue from Q1 to Q4 (Table VII) — a \$1,232 gain from lowest to highest CVR quartile — confirms that conversion efficiency is a strong standalone revenue driver beyond its interaction with spend.

G. RQ7 — Discount Effect

Table VIII and Figure 11 show the mean revenue uplift from discounts by product category. Discounts are strongly posi-

TABLE VIII: Mean Revenue by Discount Status and Category (RQ7)

Category	No Disc. (\$)	Disc. (\$)	Δ %
Electronics	10,233	13,495	+31.9%
Fashion	8,275	10,165	+22.8%
Financial Svcs	9,665	9,702	+0.4%
Home & Living	8,239	7,552	-8.3%
Sports	8,426	10,286	+22.1%

Figure 11: Effect of Discount on Revenue by Product Category (RQ7)

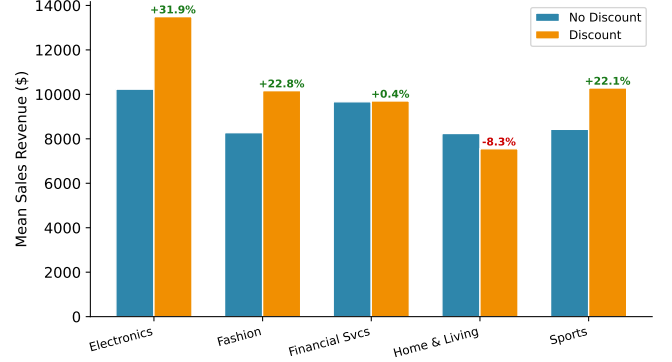


Fig. 11: Mean revenue by discount status and product category (RQ7). Percentage deltas shown above discount bars.

Figure 12: Discount Effect on Revenue by Region (RQ7)

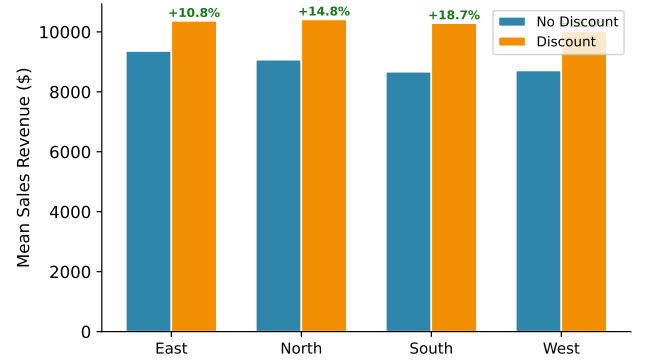


Fig. 12: Mean revenue by discount status and region (RQ7).

tive for Electronics (+31.9%), Sports (+22.1%), and Fashion (+22.8%), reflecting high price sensitivity and conversion elasticity in these categories. In contrast, Home & Living shows a slight revenue decline (−8.3%), possibly because discounts attract low-margin orders without generating proportional unit volume increases.

Figure 12 presents the regional breakdown. The discount lift is largest in the North (+25%) and smallest in the East (+8%), suggesting that Eastern markets may be less price-sensitive or that discounting cannibalises full-price sales in that region.

VIII. CONCLUSION

This paper presented a comprehensive supervised machine learning study on sales revenue prediction from marketing campaign data, covering seven research questions. The key findings are:

- 1) **Gradient Boosting** achieves the best predictive accuracy ($R^2 = 0.925$, $MAE = 1,104$), closely matched by Ridge regression, which benefits from strong linear signals in ad spend.
- 2) **Ad Spend, CVR, and CTR** are the dominant revenue predictors; demographic variables (Age, Gender) add negligible global lift.
- 3) **Diminishing returns** on ad spend are confirmed across all campaign types; Social Media saturates fastest.
- 4) **Seasonal and regional subgroups** are all highly predictable ($R^2 > 0.91$), with Summer and North being marginally superior.
- 5) **CTR and CVR amplify** the revenue impact of ad spend — high-engagement, high-spend campaigns generate disproportionately greater returns.
- 6) **Discounts** boost revenue significantly for Electronics and Sports (+22–32%) but reduce it for Home & Living (−8.3%), indicating a need for category-specific promotional strategies.

Limitations and future work: The dataset is synthetic, which may limit transferability of numerical findings. Future work should validate on real-world data, explore causal inference frameworks to distinguish correlation from true ad-spend effectiveness, and incorporate time-series features to capture campaign momentum.

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